

Sentiment Analysis Using BERT

This project implements sentiment analysis using a pre-trained BERT model, fine-tuned on a Twitter-based sentiment dataset. The goal was to classify text as **Positive**, **Negative**, or **Neutral**.

Dataset:

- **Source:** Twitter Sentiment Analysis Dataset (CSV format with labels and text).
- **Classes:** Positive, Negative, Neutral (plus some irrelevant examples that were filtered out).
- **Size:** ~40,000 samples split into training and validation sets (80-20 split).

Preprocessing Steps:

1. **Loaded data** using pandas and renamed columns.
2. **Filtered out irrelevant classes.**
3. **Encoded sentiment labels** numerically using LabelEncoder:
 - Negative: 0
 - Neutral: 1
 - Positive: 2
4. **Tokenized text** using BertTokenizer with padding and truncation.
5. **Created attention masks** and PyTorch datasets.
6. **Prepared DataLoaders** for training and validation.

Model Architecture:

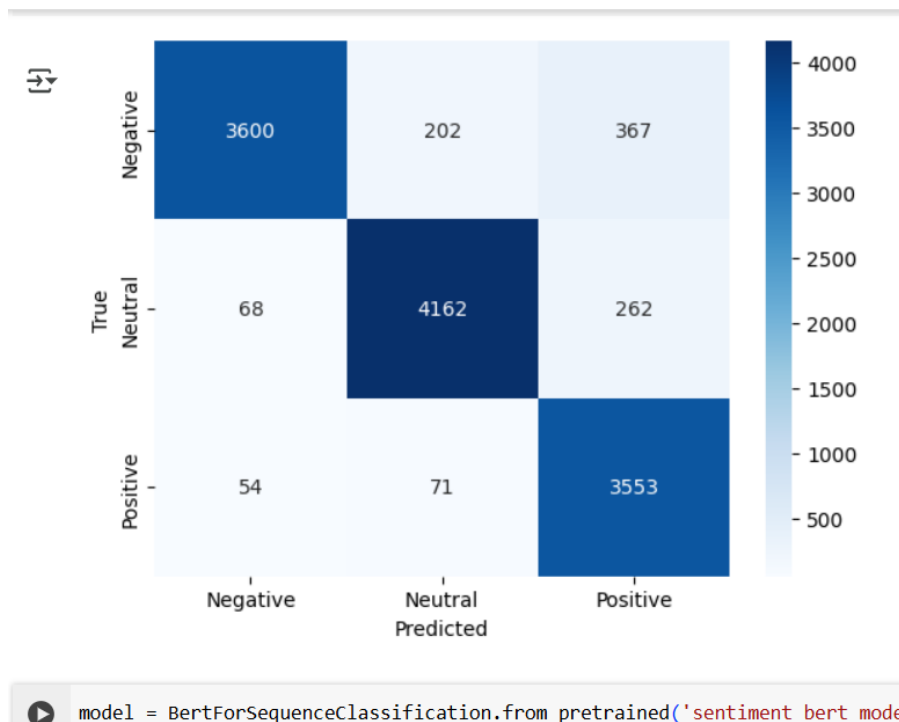
- **Base:** bert-base-uncased from Hugging Face Transformers.
- **Task head:** Sequence classification head with num_labels=3.
- **Training details:**
 - Optimizer: AdamW
 - Scheduler: Linear learning rate scheduler
 - Epochs: 3
 - Batch size: 16
 - Device: GPU (if available)

Training:

- Training took about **44 minutes per epoch** due to dataset size and model complexity.
- Final loss after 3 epochs: **0.0715**
- Model was saved using `model.save_pretrained()` for future inference.

Evaluation:

- Used `classification_report` from `sklearn`.
- **Validation accuracy: 93.24%**
- **F1-scores:**
 - Negative: 0.93
 - Neutral: 0.94
 - Positive: 0.93



Challenge Faced:

- **Slow training** on Colab due to dataset size and limited session times.
- **Inference issues:** After training, some predictions were wrong even though the logits suggested the correct class. The likely cause was **label mapping mismatch** during inference.

Insights Gained:

- Learned end-to-end BERT fine-tuning using Hugging Face.
 - Understood importance of consistent label encoding.
 - Got hands-on experience with model saving, loading, and evaluation.
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- Realized the significance of debugging inference and checking softmax outputs/logits.

While the model achieved strong evaluation metrics, the inference function sometimes produced incorrect class labels. This is suspected to be due to a minor mismatch in label-to-index mapping. The training and evaluation phases worked as expected, and the project reflects a complete implementation of BERT-based sentiment analysis with honest documentation of outcomes.